The small even factor at a critical attachment

Human Compression project: a component of the L019 recovery

The acquired large-degree reduction leaves an even transitive fractured factor of degree $d \in \{6, 8, 10\}$ and rank $d/2$, with an endpoint anchor. Here fractured means that every maximal omission is intransitive. This note supplies the exact finite boundary. Its principal application still leaves the critical symmetric factor unresolved.

Lemma 1. *No even transitive fractured string C-marking of degree* $6, 8$ *or* 10 *and rank half its degree is both imprimitive and anchored at an endpoint.*

Proof. An imprimitive transitive group preserves equal blocks. In these degrees the possible proper block sizes are respectively $2, 3$; $2, 4$; and $2, 5$. After a point relabelling its marks lie in the natural wreath action $S_k \wr S_m$, where $km = d$, and, by evenness, in

$$E = (S_k \wr S_m) \cap A_d.$$

For two blocks of odd size k, an involution interchanging the blocks consists of exactly k transpositions and is odd. Thus even involutions cannot generate a transitive group in that action. This disposes of $(d, k) = (6, 3), (10, 5)$ without a search.

The four remaining actions are exhausted by the following finite procedure. Normalize the endpoint anchor to point 1, using the transitive full wreath group $W = S_k \wr S_m$. The first mark is an even involution moving 1; all subsequent marks are even involutions fixing 1. At each step try every such involution outside the group already generated, commuting with every earlier mark except the last. Require that it not commute with the last mark. Indeed, the component of the noncommutation diagram containing the first mark is transitive: all other components commute with it and fix the anchor. They therefore fix its orbit pointwise, which is the full orbit of the marking. A nonidentity mark cannot belong to such another component. Thus the diagram is connected. Test the intersection property on every pair of subsets of the enlarged prefix. For prefixes of length at most two it follows directly from distinct nonidentity involutions.

Initially conjugate by W_1, and subsequently by its subgroup centralizing the entire chosen prefix. Try one representative of every orbit on the candidate involutions. These conjugations preserve all chosen marks, the point 1, evenness and the wreath action; they omit no possible next marking. If H is the group of a prefix of length l, reject it only when $[E : H] < 2^{d/2-l}$. Every remaining independent mark would at least double the group order, so this additional pruning is necessary. At length $d/2$ retain every transitive generated subgroup of E, without requiring it to equal E.

The code constructs W from adjacent transpositions inside each block and simultaneous transpositions swapping adjacent blocks. All subsequent subgroup and involution operations are on these literal permutation groups; no catalogue of transitive groups is used. The complete counts are as follows. A node count includes the empty prefix and then each successive prefix length; the last column counts transitive terminal tuples, before testing fracture.

d	k	$\lvert E\rvert$	nodes by prefix length	terminal tuples
6	2	24	$1, 2, 3, 0$	0
8	2	192	$1, 6, 9, 14, 0$	0
8	4	576	$1, 4, 13, 26, 4$	2
10	2	1920	$1, 6, 50, 78, 27, 0$	0

Both surviving degree-eight tuples have order 192. Omitting their last mark leaves transitive subgroups of orders 64 and 48 respectively. They are therefore not fractured. A separate literal permutation implementation enumerates these closures and their orbits, confirming the exclusion. The two odd-block cases were also run as redundant finite controls, and have no transitive terminal tuple. As a positive control, the same procedure in $S_5 \cap A_5$, at rank two, retains an anchored transitive D_{10} marking. Every recorded execution terminated with an explicit completion marker and empty standard error. A deadline or an incomplete node list would supply no part of the exclusion. □

Lemma 2. *Every anchored natural rank-five A_{10} C-marking fails the pure-leaf criterion. In particular, it cannot be an anchored factor meeting a disjointly labelled odd transitive factor at just its anchor inside a string C-marking.*

Proof. The complete acquired L044 census has eight A_{10} rank-five representatives. Its retained terminal superset includes two further tuples of order 1920; we inspect all ten, and neither additional tuple has an anchor. In each orientation of every tuple inspect every point, obtaining exactly the following eight anchored cases.

Orient the marks $t_0, \dots, t_4$ with the anchor a first. For $j \geq 1$, put $P_j = \langle t_0, \dots, t_{j-1} \rangle$, $P_0 = 1$, $X_{j-1} = a^{P_{j-1}}$, and

$$K_j = P_j \cap \mathrm{Sym}(X_{j-1}), \qquad M_j = K_j \cap P_{j-1}.$$

The symmetric group on a subset fixes its complement pointwise. The acquired leaf theorem requires $K_j \leq P_{j-1}$ for every j. Zero-based indices and orientations refer to the unchanged source list.

Index	Orientation	Anchor	j	$\lvert K_j \rvert$	$\lvert M_j \rvert$
0	1	1	3	60	10
1	0	10	4	360	60
1	1	1	3	60	10
2	0	10	3	60	10
2	1	1	4	360	60
3	0	10	3	60	10
4	1	1	4	360	60
7	0	10	4	360	60

Every row has index six, so the required inclusion fails. These are finite closure computations on the displayed prefixes, not an inference that a restricted marking has IP. A separate implementation multiplies literal permutation arrays, enumerates each finite closure, and retains exactly the permutations fixing the complement of X_{j-1}. It agrees with the GAP subgroup intersections in every row. Four cases have a nontransitive complementary tail and hence were not covered by the earlier even-attachment filter; all eight cases are included here.

If an odd transitive group on another support meets this marking only at a, the acquired odd-attachment theorem forces all these leaf inclusions by the original intersection property. The displayed failure therefore excludes such an actual attachment. □

Theorem 3. *Let an even transitive fractured string C-marking have degree $d \in \{6, 8, 10\}$, rank $d/2$, and an endpoint anchor a. Suppose it occurs in a larger C-marking together with a disjointly labelled odd transitive factor whose support meets its own exactly at a. Then it is a primitive degree-six rank-three A_5 marking, and its tail away from a has order ten and is transitive on the other five points.*

Proof. Lemma 1 excludes imprimitive actions. The complete primitive classification acquired in L047 applies at rank at least half the degree, except for the natural alternating and symmetric

groups. The latter are impossible by evenness. The complete L044 small maxima exclude natural A_6 and A_8, and Lemma 2 excludes natural A_{10} with this attachment.

The twelve retained proper-primitive representatives and their two orientations were already inspected in the faithful-augmentation component. We now impose only evenness and the endpoint anchor, without its character test or an assumed tail orbit. The literal filter leaves exactly source index/orientation pairs

$$(0,1),(1,0),(1,1),(2,0).$$

Each has degree six, rank three, group order sixty and tail order ten. Its tail fixes a and has one orbit of size five. The degree-ten representative has no anchor in either orientation. Independent literal closures reproduce the group and tail orders and verify the exact leaf inclusions for all three prefixes of the four surviving cases. Thus none is silently excluded, and these are exactly the remaining possibilities. This proves the theorem. □

Use, attribution and precise remaining scope. Combined with the separately frozen conditional overlap reduction, Theorem 3 leaves only a natural symmetric factor of even degree m and rank $m/2+1$, meeting such a six-point A_5 factor at their common endpoint anchor. The whole degree is $m+5$ and the rank is $(m+8)/2$. This is still a reduction: validity of the symmetric factor's leaf and the full stability bijection remain open within this recovery.

The complete primitive and alternating classifications and their literal lists remain attributed to Cameron–Fernandes–Leemans–Mixer through L047 and L044. The new boundary enumeration searches anchored equality ranks not covered by the older above-half imprimitive census; the eight-anchor filter expands the previous four-case attachment test. All exact acquired premises and execution inputs are pinned. No whole L019, new conjecture, Lean theorem or compression measurement is claimed.